

Theories of Computation: Summative Assignment 3

To be handed in on Canvas before **Tuesday 25th April, 12pm**

An *undirected graph* G consists of vertices and edges, and its *size* is the number of vertices. For example, here are two undirected graphs of size 4:



We take the vertices to be numbered $0, 1, \dots, n-1$, where n is the size of the graph.

A *clique* in G is a set of vertices with any two distinct vertices appearing in the list are related by an edge. The CLIQUE problem consists in deciding whether a given undirected graph G contains a clique of a given size. For example, G_1 contains a clique of length 3 but G_2 does not.

A *propositional formula* φ is constructed from atoms and the connectives $\neg$, $\vee$ and $\wedge$. The *size* $|\varphi|$ of a formula is the number of occurrences of atoms in it. An *assignment* gives value `True` or `False` to each atom; it is *satisfying* for a formula φ if it makes the formula `True`. The SAT problem consists in deciding whether a given formula φ has a satisfying assignment.

The purpose of this exercise is to establish a polytime reduction from CLIQUE to SAT.

1. Let's first consider one instance: deciding whether G_1 above contains a clique of size 3. We'll define a propositional formula $\varphi_{G_1}^3 = \alpha_{G_1}^3 \wedge \beta_{G_1}^3$ that has a satisfying assignment if and only if G_1 has a clique of size 3. (It does, but imagine that we do not yet know this.)

Such a clique, if it exists, can be listed in increasing order. For $i < 3$ and $p < 4$, the atom $x_{i,p}$ means that p is the i th element of the list. This forms our set $At = \{x_{0,0}, \dots, x_{0,3}, x_{1,0}, \dots, x_{1,3}, x_{2,0}, \dots, x_{2,3}\}$ of propositional atoms.

The first part of the formula will be

$$\begin{aligned} \alpha_{G_1}^3 = & (x_{0,0} \vee x_{0,1} \vee x_{0,2} \vee x_{0,3}) \wedge (x_{1,0} \vee x_{1,1} \vee x_{1,2} \vee x_{1,3}) \wedge (x_{2,0} \vee x_{2,1} \vee x_{2,2} \vee x_{2,3}) \wedge \\ & \neg((x_{0,0} \wedge x_{0,1}) \vee (x_{0,0} \wedge x_{0,2}) \vee (x_{0,0} \wedge x_{0,3}) \vee (x_{0,1} \wedge x_{0,2}) \vee (x_{0,1} \wedge x_{0,3}) \vee (x_{0,2} \wedge x_{0,3})) \wedge \\ & \neg((x_{1,0} \wedge x_{1,1}) \vee (x_{1,0} \wedge x_{1,2}) \vee (x_{1,0} \wedge x_{1,3}) \vee (x_{1,1} \wedge x_{1,2}) \vee (x_{1,1} \wedge x_{1,3}) \vee (x_{1,2} \wedge x_{1,3})) \wedge \\ & \neg((x_{2,0} \wedge x_{2,1}) \vee (x_{2,0} \wedge x_{2,2}) \vee (x_{2,0} \wedge x_{2,3}) \vee (x_{2,1} \wedge x_{2,2}) \vee (x_{2,1} \wedge x_{2,3}) \vee (x_{2,2} \wedge x_{2,3})) \end{aligned}$$

to express having a list of 3 vertices of G_1 .

Write a formula $\beta_{G_1}^3$, shorter than $\alpha_{G_1}^3$, to express that the list is in increasing order and represents a clique.

Hint: First describe the relationship between the 0th and 1st element of the list. For example, for G_2 above, it would be $(x_{0,0} \wedge x_{1,2}) \vee (x_{0,1} \wedge x_{1,2}) \vee (x_{0,1} \wedge x_{1,3})$. There is no need to use negation in your formula.

[3 marks]

2. Now we want to solve the general problem: for a graph G of size n , does it have a clique of given size $k \leq n$?

To do this let us extend the definition in part 1 to a formula $\varphi_G^k = \alpha_G^k \wedge \beta_G^k$ that has a satisfying assignment if and only if G has a clique of size k . Again, if a clique exists, it can be listed in increasing order. For $i < k$ and $p < n$, the atom $x_{i,p}$ means that p is the i th element of the list. To begin, we write

$$\alpha_G^k = \bigwedge_{i < k} \left(\bigvee_{p < n} x_{i,p} \right) \wedge \bigwedge_{i < k} \neg \left(\bigvee_{p < q < n} (x_{i,p} \wedge x_{i,q}) \right)$$

to express having a list of k vertices of G .

Give a formula β_G^k to express that the list is in increasing order and represents a clique. The set of edges is written E , so you use the condition $(p, q) \in E$ to indicate that there is an edge from p to q . **[3 marks]**

3. Show that the size of φ_G^k is polynomially bounded by the size of G , that is, there exists a polynomial P such that $|\varphi_G^k| \leq P(n)$.

Hint: You can find polynomial bounds for $|\alpha_G^k|$ and for $|\beta_G^k|$ independently. This means that you can partially answer this question even if you could not answer the previous ones. **[3 marks]**

4. It follows that the time taken to construct φ_G^k from G is polynomial in the size of G (since we can check in linear time whether $(p, q) \in E$). Using what you know about the SAT problem, explain in a few lines what you could deduce about the CLIQUE problem if it were the case that $\mathcal{P} = \mathcal{NP}$. **[3 marks]**

Solution 1 1.

$$\begin{aligned} \beta_{G_1}^3 = & [(x_{0,0} \wedge x_{1,1}) \vee (x_{0,0} \wedge x_{1,2}) \vee (x_{0,1} \wedge x_{1,2}) \vee (x_{0,1} \wedge x_{1,3})] \wedge \\ & [(x_{0,0} \wedge x_{2,1}) \vee (x_{0,0} \wedge x_{2,2}) \vee (x_{0,1} \wedge x_{2,2}) \vee (x_{0,1} \wedge x_{2,3})] \wedge \\ & [(x_{1,0} \wedge x_{2,1}) \vee (x_{1,0} \wedge x_{2,2}) \vee (x_{1,1} \wedge x_{2,2}) \vee (x_{1,1} \wedge x_{2,3})] \end{aligned}$$

2.

$$\beta_G^k = \bigwedge_{i < j < k} \bigvee_{\substack{p < q < n \\ (p,q) \in E}} (x_{i,p} \wedge x_{j,q})$$

3. The first part of α_G^k is $\bigwedge_{i < k} \left(\bigvee_{p < n} x_{i,p} \right)$ which has size at most kn . The second part is $\bigwedge_{i < k} \neg \left(\bigvee_{p < q < n} (x_{i,p} \wedge x_{i,q}) \right)$ which has size at most $2kn^2$. The formula β_G^k has size at most $2k^2n^2$. Thus the size of φ_G^k is at most $kn + 2kn^2 + 2k^2n^2$, which is at most $n^2 + 2n^3 + 2n^4$ since $k \leq n$.
4. Since SAT is in $\mathcal{NP}$, and we have reduced CLIQUE to SAT in polytime, it follows that CLIQUE is in $\mathcal{NP}$. So if $\mathcal{P} = \mathcal{NP}$, then CLIQUE can be solved in polynomial time.